KLE Technological University | Creating Value, Leveraging Knowledge

UNIT-I

Chapter 1: Introduction

- Overview of Big Data
- Big Data Characteristics
- Different Types of Data
- Data Analytics
- Data Analytics Life Cycle

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Overview of Big Data

Data refers to raw facts, figures, observations, or information collected from different sources.

Data can be in the form of:

- Numbers
- Text
- Images
- Audio
- Video
- Documents
- Transactions
- Sensor readings
- Social media posts

Example: Suppose a college collects information about its students:

Student ID	Name	Age	Marks	Attendance
101	Anu	19	82	91%
102	Ravi	20	75	86%
103	Priya	19	91	94%

The values stored in this table are data.

Big Data is data that is too large, too fast, or too diverse to be handled efficiently using traditional data-management techniques.

- The important point is that **Big Data is not simply "a large amount of data."**
- According to the Gartner definition the Big Data is **"High-volume, high-velocity, or high-variety information assets that require new forms of processing."**

It may be challenging because of:

- **Huge volume** of data
- **High speed** of data generation
- **Different types and formats** of data
- **Uncertain or unreliable data**
- Difficulty in extracting **useful value**

Example: Social Media - Consider a platform such as Instagram.

Every second, users will generate the data such as:

- Posts
- Photos

- Videos
- Comments
- Likes
- Shares
- Messages
- Search activity
- Location-related information
- User interaction logs

This data is: **Large in volume + generated continuously + available in many formats.**

Therefore, social-media platforms need Big Data technologies to store and process such information efficiently.

Why Did Big Data Become Important?

The amount of digital data is being generated has increased enormously because of the growth of technology. Such as Social Media, Smart Phones, Internet of Things (IoT), Business Transactions, Machines and Sensors, etc.

Traditional Data vs Big Data

Traditional Data	Big Data
• Usually smaller in volume	• Extremely large volume
• Mostly structured	• Structured, semi-structured and unstructured
• Often generated at a relatively manageable rate	• Can be generated continuously at very high speed
• Usually handled by traditional databases	• Often requires distributed storage and processing
• Processing can often be performed on a single/centralized system	• Frequently requires distributed systems
• Examples: employee records, bank tables	• Social media, IoT, sensor, multimedia and large-scale transaction data

Why Traditional Systems Face Problems with Big Data?

Suppose an organization has only **10,000 customer records**. A traditional database can usually manage this easily.

Now imagine: **100 million customers + millions of transactions every day + images + videos + logs + sensor data.**

The system has to answer questions such as:

- Where should the data be stored?
- How can storage capacity be increased?
- How can data be processed faster?
- How can failures be handled?
- How can different data formats be managed?
- How can useful information be extracted?

This is where **Big Data technologies and distributed computing approaches** become important.

Applications of Big Data

Domain	Example Use
1. **Healthcare**	Analysis of patient and medical data
2. **Banking**	Fraud detection
3. **Retail**	Customer behavior and recommendations
4. **Transportation**	Route optimization
5. **Manufacturing**	Predictive maintenance
6. **Education**	Student performance analysis
7. **Government**	Smart-city services
8. **Social Media**	Recommendation and trend analysis
9. **Telecommunications**	Network and customer analysis
10. **Energy**	Monitoring consumption and demand

Big Data Characteristics

Big Data is not defined only by its size. It has several important characteristics that make it different from traditional data.

The most commonly discussed characteristics are the **5 V's of Big Data**:

Volume + Velocity + Variety + Veracity + Value

1. Volume: Volume refers to the enormous quantity of data generated and collected from various sources. In Big Data, the amount of data can be so large that traditional storage systems may not be sufficient or efficient.

Volume = How much data is being generated?

Example: College
Imagine your college maintains data for thousands of students.

Data may include:

- Admission records
- Attendance
- Examination marks
- Assignments
- Library transactions
- Fee transactions
- LMS activity
- Project records

If this information is collected year after year, the total volume becomes very large.

2. Velocity: Velocity refers to the speed at which data is generated, collected, transmitted and processed.

Velocity = How fast is the data being generated and processed?

Example: Online Payments

Suppose you make a UPI payment. The system needs to process information such as:

Payment initiated
↓
Transaction verified
↓
Amount transferred
↓
Transaction recorded
↓
Confirmation received

This happens within seconds. If the system takes several hours to process the transaction, it would not be useful to the customer. Therefore, the **speed of data processing matters**.

3. Variety: Variety refers to the different types, formats and sources of data generated in Big Data environments. It contains text, symbols, numbers, audio, images, videos, documents, sensor data, logs, etc.

Variety = What different forms of data are we dealing with?

Example: YouTube

A single platform handles:

Text: Titles, descriptions, comments

Images: Thumbnails

Video: Uploaded videos

Audio: Soundtracks and speech

Numbers: Views, likes, ratings

Logs: User activity

So YouTube deals with a **wide variety of data**.

4. Veracity: Veracity refers to the reliability, accuracy and trustworthiness of data. Not all data collected by an organization is necessarily correct or useful.

Veracity = Can we trust the data?

Example: Customer Reviews
Suppose an online shopping website receives 1 million reviews.
Among them:

- Genuine reviews
- Duplicate reviews
- Fake reviews
- Spam
- Irrelevant comments

If the company analyzes all of them without checking their reliability, it may get incorrect conclusions. Therefore, the data needs to be evaluated and cleaned.

5. Value: Value refers to the useful information or insights that can be extracted from Big Data. Simply collecting huge amounts of data is not enough.

The data must help an organization:

- Understand a situation

- Discover patterns
- Make decisions
- Improve processes
- Solve problems
- Create business benefits

Value = What useful information can we obtain from the data?

Example: Online Shopping

Suppose an online shopping company stores: **10 TB of customer data.**

Is that automatically valuable? **No.**

The company needs to analyze the data.

For example, it may discover: "Customers who purchase a mobile phone frequently purchase a mobile cover within the next two days."

This is a **useful insight**. The company can use it to recommend mobile covers to customers.

Therefore: **Data → Analysis → Insight → Value**

Different Types of Data

In Big Data environments, data does not always come in the form of neat tables. It can have different structures and formats.

1. Structured Data: Structured data is data that has a well-defined structure or schema and is organized in a consistent format, usually in rows and columns. It can be easily stored, searched, queried and processed using traditional relational database systems.

Structured Data = Data arranged in a fixed and organized format.

Example: Consider a student database:

Student_ID	Name	Age	Marks
101	Anu	19	85
102	Ravi	20	78
103	Priya	19	92

Here:

- Each row represents one student.
- Each column represents one attribute.
- The structure is clearly defined.
- Data follows a consistent format.

Therefore, this is **structured data**.

2. Semi-Structured Data: Semi-structured data is data that does not follow a rigid tabular structure like a relational database, but still contains some organizational elements, such as tags, keys, attributes or metadata.

Semi-Structured Data = Data that has some organization, but does not follow a fixed table structure.

Example: Consider JSON

```
{
  "student_id": 101,
  "name": "Anu",
  "marks": 85
}
```

Here the data is organized using:

- Keys
- Values
- Nested structures

But it is not arranged as a traditional relational table. Therefore, JSON is a common example of semi-structured data.

3. Unstructured Data: Unstructured data is data that does not have a predefined or fixed structure and does not follow the formal organization of relational data models.

Unstructured Data = Data without a predefined tabular structure.

Example: A photograph, An audio recording, A YouTube video are not naturally organized into rows and columns.

Structured vs Semi-Structured vs Unstructured

Feature	Structured	Semi-Structured	Unstructured
Structure	Well-defined	Partially defined	No predefined structure
Schema	Fixed	Flexible	No fixed schema
Organization	Rows and columns	Tags, keys, attributes, etc.	No fixed organization
Format	Tabular	Flexible	Various formats

Feature	Structured	Semi-Structured	Unstructured
Examples	Relational database tables	JSON, XML	Images, audio, video
Querying	Relatively easy	Requires suitable tools	More complex
Flexibility	Low	High	Very high
Example	Banking transactions	Web/API data	Social media

Data Analytics

In today's digital world, organizations generate huge amounts of data from different sources such as:

- Websites
- Mobile applications
- Social media
- Online transactions
- IoT devices
- Sensors
- Customer interactions
- Business systems

Simply collecting this data is not enough. The organization needs to examine and analyze the data to discover useful information and make better decisions. This process is known as Data Analytics.

Definition of Data Analytics:

Data Analytics is the process of examining, cleaning, transforming, and analyzing data to discover useful information, patterns, trends, relationships, and insights that support decision-making.

Example: Suppose a college has student data containing:

- Attendance
- Internal assessment marks
- Assignment marks
- Semester marks
- Previous academic performance

The college has the data. But simply storing the data doesn't tell us much.

Using Data Analytics, we can analyze it and discover: Students with low attendance are more likely to have lower academic performance.

This is an **insight obtained from data**. The college can then take appropriate action, such as identifying students who need academic support.

Why is Data Analytics Needed?

Organizations generate enormous amounts of data every day. Without analysis, the data remains largely unused.

Data Analytics helps organizations such as:

1. Understand Data: It helps identify what is happening within the organization.
2. Discover Patterns: It can reveal repeated patterns in data.
3. Identify Trends: It helps determine how something changes over time.
4. Support Decision-Making: Organizations can use data-based insights to make informed decisions.
5. Identify Problems: Analysis can reveal unusual behavior, errors, or potential problems.
6. Improve Performance: Organizations can identify areas that need improvement.

Data Analytics vs Data Analysis

- Data Analysis = Examining the data
- Data Analytics = Broader process of working with data to obtain insights

Types of Data Analytics

1. Descriptive Analytics (What Happened?):

Descriptive Analytics focuses on understanding what has happened or what is happening by analyzing historical and current data.

Example: College

Suppose the college analyzes semester results.

Pass Percentage = 87%

Average Marks = 72%

Highest Marks = 96%

This tells us **what happened**. It does not necessarily explain why students obtained those results. That is **Descriptive Analytics**.

2. Diagnostic Analytics (Why did it Happen?):

Diagnostic Analytics focuses on finding the reasons or factors behind an event or result.

Example: College

Suppose descriptive analytics shows: "Student pass percentage decreased from 90% to 78%."

Diagnostic analytics asks: **Why did the pass percentage decrease?**

Further analysis might reveal:

- One particular subject had lower performance.
- Attendance decreased.
- Students performed poorly in a particular unit.

Now we are trying to understand the **reason behind the result**.

3. Predictive Analytics (What is likely to Happen?):

Predictive Analytics uses historical data, statistical techniques and machine-learning approaches to estimate what is likely to happen in the future.

Example: Online Shopping

Suppose an e-commerce company analyzes:

- Previous purchases
- Search history
- Customer preferences
- Seasonal trends

It may predict: "Demand for umbrellas is likely to increase during the rainy season." The company can prepare inventory accordingly.

4. Prescriptive Analytics (What should we Do?):

Prescriptive Analytics goes one step beyond prediction. It recommends what action should be taken to achieve a desired outcome.

Example: Food Delivery

Suppose analytics predicts: "Demand will be very high in Area A between 7 PM and 9 PM."

Prescriptive analytics may recommend: "Position more delivery partners in Area A before 7 PM."

Here: **Prediction → Action recommendation**

One Example to Understand All Four Types

Let's use a **college attendance example**.

Descriptive Analytics - What happened?

"Class attendance decreased from 88% to 72%."

Diagnostic Analytics - Why did it happen?

"Attendance decreased mainly because students were absent during a particular period."

Predictive Analytics - What is likely to happen?

"Students with consistently low attendance are likely to face attendance shortages."

Prescriptive Analytics - What should we do?

"Identify students with low attendance and conduct timely follow-up."

Type	Main Question	Purpose	Example
Descriptive	What happened?	Summarize past/current data	Pass percentage report
Diagnostic	Why did it happen?	Find causes	Analyze reasons for low pass percentage
Predictive	What is likely to happen?	Forecast future outcomes	Predict future student performance
Prescriptive	What should we do?	Recommend actions	Suggest intervention for at-risk students

Data Analytics Life Cycle

Data Analytics Life Cycle is a systematic sequence of phases followed to identify a problem, collect and prepare relevant data, analyze the data, communicate the findings, and use the results for decision-making.

Why Do We Need a Data Analytics Life Cycle?

Suppose the Head of a department asks: **"Why are some students performing poorly in a particular course?"**

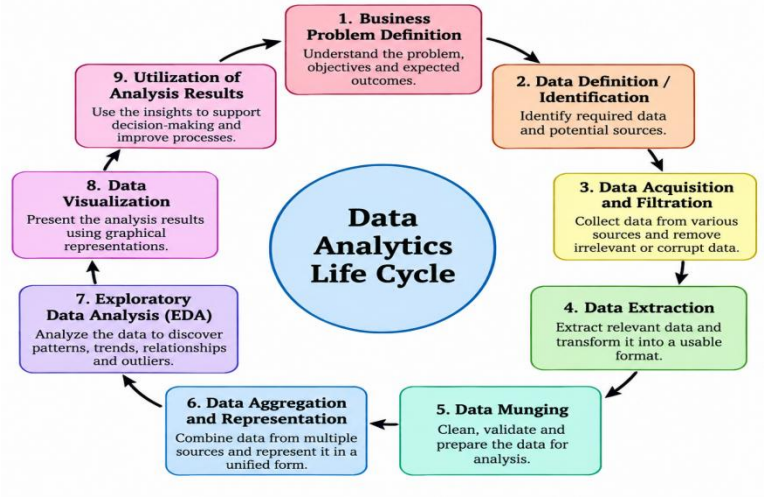
We cannot immediately start to analyze the random data. First, we need to:

1. Understand the problem.
2. Identify what data is required.
3. Collect the data.
4. Clean the data.
5. Combine relevant data.
6. Analyze it.
7. Visualize the findings.
8. Take appropriate action.

This complete process is the **Data Analytics Life Cycle**.

Phases of Data Analytics Life Cycle

1. Business Problem Definition
2. Data Definition / Identification
3. Data Acquisition and Filtration
4. Data Extraction
5. Data Munging
6. Data Aggregation and Representation
7. Exploratory Data Analysis (EDA)
8. Data Visualization
9. Utilization of Analysis Results



Phase I – Business Problem Definition:
Business Problem Definition is the first phase of the Data Analytics Life Cycle. In this phase, the organization clearly identifies and defines the problem that needs to be solved using data. The problem should be clearly stated rather than being expressed in a very general manner.

Phase II – Data Definition / Identification:
Once the problem has been clearly defined, the next step is to identify what data is required to solve the problem and where that data can be obtained. This phase involves identifying appropriate datasets and potential data sources.

Phase III – Data Acquisition and Filtration:
Data Acquisition is the process of collecting the required data from the identified sources. After collecting the data, it may contain irrelevant, incomplete or corrupted records. Therefore, filtration is performed to retain the data that is relevant to the problem.

Phase IV – Data Extraction:
Data Extraction is the process of obtaining the specific data required for analysis from the collected datasets and converting incompatible information into a suitable form when necessary. Not every piece of information in a dataset is useful for the current problem. Therefore, the relevant attributes and records are extracted.

Phase V – Data Munging:
Data Munging is the process of cleaning, transforming and validating raw data so that it becomes suitable for analysis.

Phase VI – Data Aggregation and Representation:
Data Aggregation is the process of combining data from multiple sources into a unified dataset. Organizations often store different types of information in different systems. To perform meaningful analysis, related data needs to be brought together.

Phase VII – Exploratory Data Analysis (EDA):
Exploratory Data Analysis (EDA) is the process of examining and exploring the prepared dataset to discover patterns, trends, relationships, unusual observations and important characteristics of the data. This is where analysts begin to understand what the data is actually telling them.

Phase VIII – Data Visualization:
Data Visualization is the process of representing data and analytical results using graphs, charts, plots, dashboards or other visual forms so that information can be understood easily. Large datasets can be difficult to understand when presented only as numbers or tables. Visualization makes important patterns and trends easier to identify.

Phase IX – Utilization of Analysis Results:
This is the final phase of the Data Analytics Life Cycle. Utilization of Analysis Results means using the insights obtained from analysis to support decision-making, improve processes, solve problems and take appropriate actions. The purpose of analytics is not merely to create charts or discover patterns.

Example: Analyzing Student Performance

Problem: The college wants to find out why some students are scoring low in a particular subject.

Phase	Explanation using the same example
1. Business Problem Definition	The college identifies the problem: some students are scoring low in a particular subject. The objective is to find the factors that may be associated with their low performance.
2. Data Definition / Identification	The college decides what data is needed, such as student attendance, ISA marks, assignment marks, practical marks and final examination marks.
3. Data Acquisition and Filtration	The required data is collected from the college academic system and LMS. Records that are irrelevant, such as students from another semester, are removed.
4. Data Extraction	Only the required information is extracted from the collected data, such as Student ID, attendance, ISA marks, assignment marks and examination marks.
5. Data Munging	The extracted data is cleaned. Missing values, duplicate records and incorrect entries are identified and handled so that the data is suitable for analysis.
6. Data Aggregation and Representation	Data from different sources, such as attendance and examination records, is combined using Student ID to create one unified dataset.
7. Exploratory Data Analysis (EDA)	The combined data is analyzed to discover patterns and relationships. For example, the analysis may show that students with low attendance generally have lower marks.
8. Data Visualization	The findings are presented using charts or graphs, such as a graph comparing attendance percentage with examination marks. This makes the pattern easier to understand.
9. Utilization of Analysis Results	The college uses the findings to identify students who need academic support, monitor their attendance and provide appropriate guidance or remedial support.

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THANK-YOU

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